Final Assignment – Linux OS by Lior Tzlil Friedman

Part 1: Users, Permissions & Ownership

Part 1: Users, Permissions & Ownership (15 pts)

Tasks:

1. Create **3 users**: alice, bob, and guest.
2. Create a group named developers and add both alice and bob to it.
3. Set the following permission structure:
   - A shared folder /devproject that only the developers group can read/write.
   - A file /devproject/specs.txt readable only by alice.
4. Change ownership of /devproject to user bob and group developers.

Deliverables:

- Command outputs of:
  - cat /etc/passwd | grep alice
  - ls -ld /devproject
  - ls -l /devproject/specs.txt

---

1. Create 3 users: alice, bob, and guest:

```
alice:x:1009:1013::/home/alice:/bin/bash
bob:x:1010:1014::/home/bob:/bin/bash
guest:x:1011:1015::/home/guest:/bin/bash
```

2. Create a group named Developers and add both alice and bob to it

```
Developers:x:1016:alice,bob
```

3. Set the following permission structure:

* A shared folder DevProject that only the developers group can read/write

```
┌──(lior5568㉿Lior5568)-[~]
└─$ sudo ls -la ~/DevProject
[sudo] password for lior5568:
total 8
drwxrwx---  2 bob      Developers 4096 Jul  1 03:03 .
```

* A file /DevProject/specs.txt readable only by alice

```
┌──(lior5568㉿Lior5568)-[~]
└─$ sudo ls -l ~/DevProject/specs.txt
-r-------- 1 alice alice 0 Jul  1 03:03 /home/lior5568/DevProject/specs.txt
```

4. Change ownership of /DevProject to user bob and group Developers.

```
┌──(lior5568㉿Lior5568)-[~]
└─$ sudo ls -la ~/DevProject
[sudo] password for lior5568:
total 8
drwxrwx---  2 bob      Developers 4096 Jul  1 03:03 .
```

Part 2: System Optimization Basic

🚀 **Part 2: System Optimization Basics (10 pts)**

Tasks:

1. Use top, htop, or vmstat to monitor system resource usage.
2. Write a short report on:
   - How to identify a CPU-hungry process.
   - How to stop or renice it.
3. Add a cron job to clear /tmp every day at midnight.

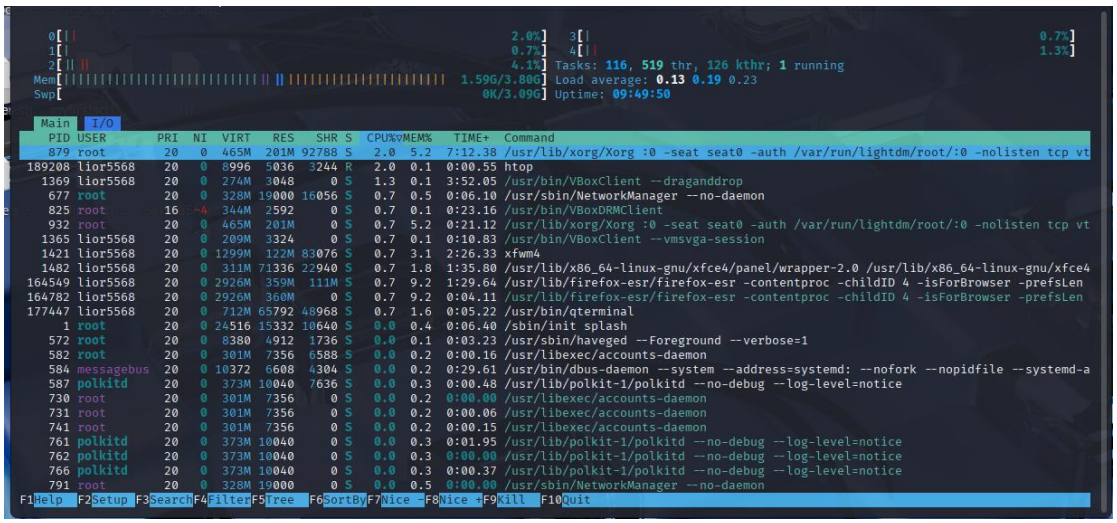
📤 **Deliverables:**

- Screenshot of command output (top or htop).
- Crontab entry with explanation.

---

1. Use top, htop, or vmstat to monitor system resource usage.
I used htop

Screenshot of the commend "htop" :



2. Write a short report on :
* How to identify a CPU-hungry process.
* How to stop or renice it

To identify a CPU-hungry or high usage process we simply can see it on the on the CPU section in the "htop" commend the higher number the higher the CPU-hunger/use.

screenshot of the CPU in commend "htop":

The CPU process can be stop using PID (Process ID)
To terminate it gracefully: kill
in htop we entering the commend: "kill (and Enter) " and then all the running process stops.

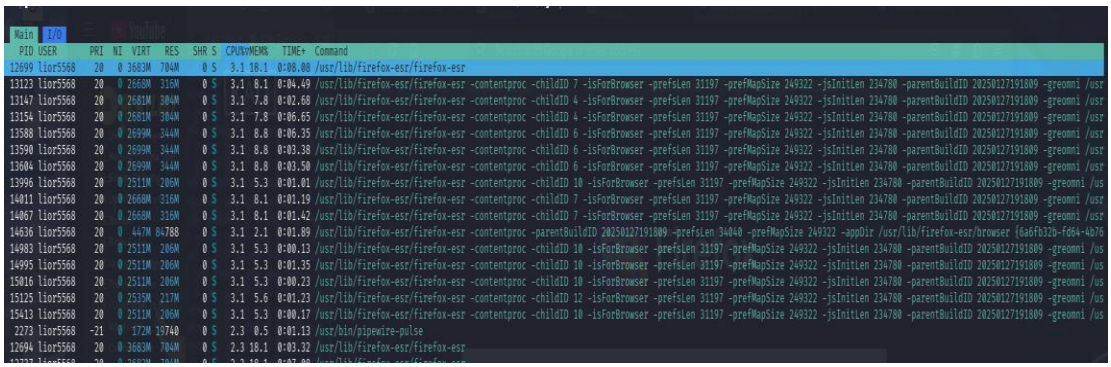
If you want to reduce the priority of a process instead of killing it, you can use the `renice` command:

Example :
renice +10 -p <PID>

Lower priority values mean higher priority (more CPU time).

Higher positive values reduce the process's CPU share.

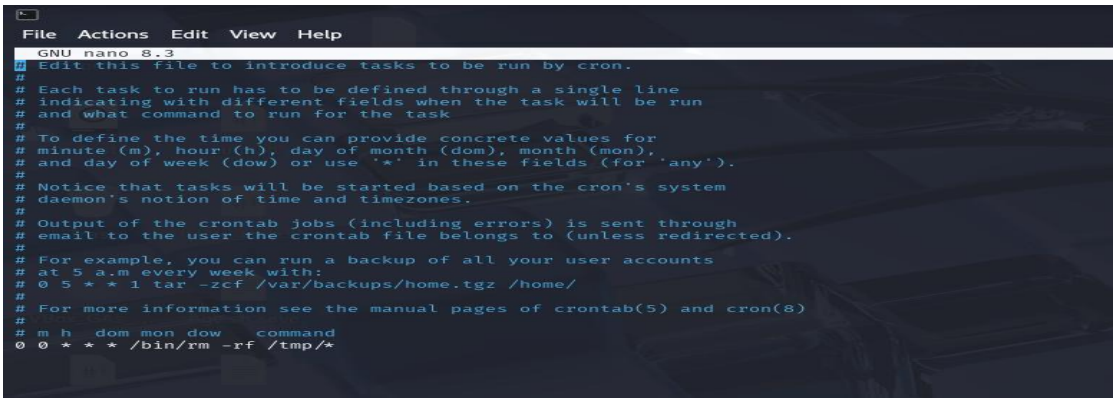
Example to increase priority:
renice -5 -p <PID>



3. Add a cron job to clear /tmp every day at midnight.
*Corn Job – It is a service in Linux systems that runs automatic tasks at predetermined times.

This cron job runs daily at midnight (00:00) and deletes all files and subdirectories inside the /tmp folder using the command: " 0 0 * * * /bin/rm –rm /tmp/* "

Explanation : 0 – zero minutes , 0 – zero hours , * - every day , * - every mouth ,
* - every day of the week , bin – folder bin in our kali linux , rm – deleting files ,
rf – deleting folders and there contains by force , /tmp/* - everting inside the folder tmp.



Part 3: Package Managers – apt and dpkg

Part 3: Package Managers – apt and dpkg (10 pts)

Tasks:

1. Install apache2, vsftpd, and samba using apt.
2. Use dpkg to verify the installation of any one of the packages.
3. Remove telnet if installed.

Deliverables:

- Output of:
  - dpkg -l | grep apache2
  - apt list --installed | grep samba
  - sudo apt remove telnet

2. Insulation of apache2

```
┌──(lior5568㉿Lior5568)-[~]
└─$ dpkg -l | grep apache2
ii  apache2                    2.4.63-1        amd64    Apache HTTP Server
ii  apache2-bin                2.4.63-1        amd64    Apache HTTP Server
ii  apache2-data               2.4.63-1        all      Apache HTTP Server
ii  apache2-utils              2.4.63-1        amd64    Apache HTTP Server
ii  libapache2-mod-php         2:8.4+96        all      server-side, HTML-e
ii  libapache2-mod-php8.4      8.4.4-1         amd64    server-side, HTML-e

┌──(lior5568㉿Lior5568)-[~]
└─$
```

```
amd64        Apache HTTP Server
amd64        Apache HTTP Server (modules and other binary files)
all          Apache HTTP Server (common files)
amd64        Apache HTTP Server (utility programs for web servers)
all          server-side, HTML-embedded scripting language (Apache 2 module) (default)
amd64        server-side, HTML-embedded scripting language (Apache 2 module)
```

Insulation of samba

```
┌──(lior5568㉿Lior5568)-[~]
└─$ apt list --installed | grep samba

WARNING: apt does not have a stable CLI interface. Use with caution in scripts.

libldb2/kali-rolling,now 2:2.11.0+samba4.22.2+dfsg-1 amd64 [installed,automatic]
libtalloc2/kali-rolling,now 2:2.4.3+samba4.22.2+dfsg-1 amd64 [installed,automatic]
libtdb1/kali-rolling,now 2:1.4.13+samba4.22.2+dfsg-1 amd64 [installed,automatic]
libtevent0t64/kali-rolling,now 2:0.16.2+samba4.22.2+dfsg-1 amd64 [installed,automatic]
python3-ldb/kali-rolling,now 2:2.11.0+samba4.22.2+dfsg-1 amd64 [installed,automatic]
python3-samba/kali-rolling,now 2:4.22.2+dfsg-1 amd64 [installed,automatic]
python3-talloc/kali-rolling,now 2:2.4.3+samba4.22.2+dfsg-1 amd64 [installed,automatic]
python3-tdb/kali-rolling,now 2:1.4.13+samba4.22.2+dfsg-1 amd64 [installed,automatic]
samba-ad-dc/kali-rolling,now 2:4.22.2+dfsg-1 amd64 [installed,automatic]
samba-ad-provision/kali-rolling,now 2:4.22.2+dfsg-1 all [installed,automatic]
samba-common-bin/kali-rolling,now 2:4.22.2+dfsg-1 amd64 [installed,automatic]
samba-common/kali-rolling,now 2:4.22.2+dfsg-1 all [installed,automatic]
samba-dsdb-modules/kali-rolling,now 2:4.22.2+dfsg-1 amd64 [installed,automatic]
samba-libs/kali-rolling,now 2:4.22.2+dfsg-1 amd64 [installed,automatic]
samba/kali-rolling,now 2:4.22.2+dfsg-1 amd64 [installed]
tdb-tools/kali-rolling,now 2:1.4.13+samba4.22.2+dfsg-1 amd64 [installed,automatic]

┌──(lior5568㉿Lior5568)-[~]
```

Part 4: File Paths, Folders & Relative Navigation

Part 4: File Paths, Folders & Relative Navigation (10 pts)

Tasks:

1. Create the following folder structure:
2. /home/student/project/
3. ├── reports/
4. ├── logs/
5. └── scripts/
6. From within scripts/, write the relative path to access reports/.
7. Write the absolute path to the logs/ folder.
8. Copy a file from reports/ to logs/ using both relative and absolute paths.

Deliverables:

- Screenshot or output of commands used.
- Explanation of when and why to use relative vs full paths.

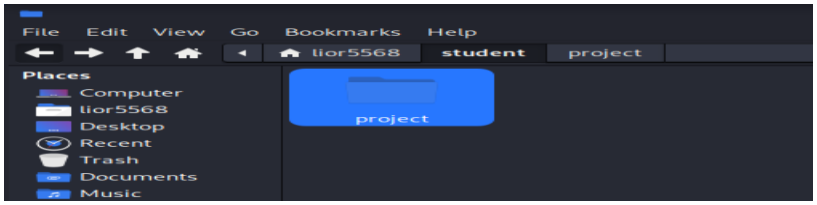
A. The output of the commands:

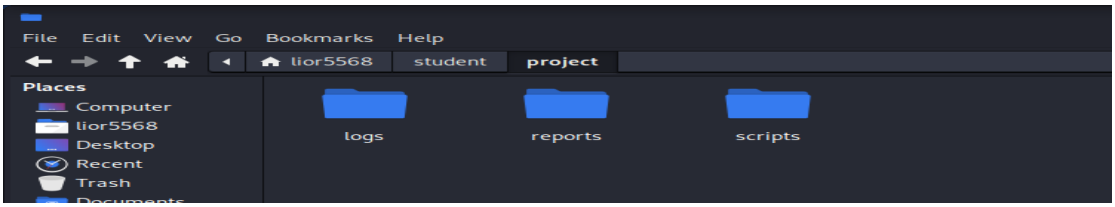
* The Folder structure

```
┌──(lior5568㉿Lior5568)-[~]
└─$ sudo mkdir /home/lior5568/project

┌──(lior5568㉿Lior5568)-[~]
└─$ sudo rm -R /home/student

┌──(lior5568㉿Lior5568)-[~]
└─$ sudo mkdir /home/lior5568/student

┌──(lior5568㉿Lior5568)-[~]
└─$ sudo mkdir /home/lior5568/student/project

┌──(lior5568㉿Lior5568)-[~]
└─$ sudo mkdir /home/lior5568/student/project/reports

┌──(lior5568㉿Lior5568)-[~]
└─$ sudo mkdir /home/lior5568/student/project/logs

┌──(lior5568㉿Lior5568)-[~]
└─$ sudo mkdir /home/lior5568/student/project/scripts
```

The Folders

```
┌──(lior5568㉿Lior5568)-[~]
└─$ ls -la /home/lior5568/student/project
total 20
drwxr-xr-x 5 root root 4096 Jul  6 04:50 .
drwxr-xr-x 3 root root 4096 Jul  6 04:49 ..
drwxr-xr-x 2 root root 4096 Jul  6 04:50 logs
drwxr-xr-x 2 root root 4096 Jul  6 04:49 reports
drwxr-xr-x 2 root root 4096 Jul  6 04:50 scripts
```





B. From within **scripts** write the relative path to access **reports**

scripts

```
┌──(lior5568㉿Lior5568)-[~/student/project/scripts]
└─$
```

From **scripts** to **reports**

```
┌──(lior5568㉿Lior5568)-[~/student/project/scripts]
└─$ ..

┌──(lior5568㉿Lior5568)-[~/student/project]
└─$ ./reports

┌──(lior5568㉿Lior5568)-[~/student/project/reports]
└─$
```

C. The absolute path to the logs/ folder

```
File  Actions  Edit  View  Help
Welcome The King lior5568
┌──(lior5568㉿Lior5568)-[~]
└─$ cd /home/lior5568/student/project/logs

┌──(lior5568㉿Lior5568)-[~/student/project/logs]
└─$ pwd
/home/lior5568/student/project/logs
```

D. Copy a file from reports/ to logs/ using both relative and absolute paths.

* A file in directory **reports**

```
┌──(lior5568㉿Lior5568)-[~/student/project/reports]
└─$ ls
reports.txt
```

copying to **logs** via **relative** path

```
┌──(lior5568㉿Lior5568)-[~/student/project/reports]
└─$ sudo cp reports.txt ../logs
[sudo] password for lior5568:
```

the file in directory **logs**

```
┌──(lior5568㉿Lior5568)-[~/student/project/logs]
└─$ ls
reports.txt
```

copying to logs via **absolute** path

```
┌──(lior5568㉿Lior5568)-[~/student/project/logs]
└─$ sudo cp /home/lior5568/student/project/reports/reports.txt /home/lior5568/student/project/
logs

┌──(lior5568㉿Lior5568)-[~/student/project/logs]
└─$ ls
reports.txt
```

Explanation of when and why to use relative vs full paths:

Relative paths are used when we want flexibility and portability
they are shorter and depend on our current working directory.
example: ../ reports means " go up one directory and then into 'reports' ".

Absolute paths always start from root (/) and are more stable
they are useful in scripts, cron jobs, and system-level configuration.
example: /home/student/project/logs is always valid , no matter where we are in the system.

Part 5: Service Configuration

Tasks:

1. Start and enable the following services using systemctl:
   - apache2
   - ssh
   - vsftpd
   - smbd (samba)
2. Use netstat or ss to verify that each service is listening.
3. Create a table listing:
   - Service name
   - Port number
   - Purpose
   - Command used to check status

📥 **Deliverables:**

- Screenshot of systemctl status for each service.
- Table of services, ports, and status verification.

systemctl status of Smbd (Samba):

```
┌──(lior5568㉿Lior5568)-[~]
└─$ sudo systemctl status smbd
● smbd.service - Samba SMB Daemon
     Loaded: loaded (/usr/lib/systemd/system/smbd.service; enabled; preset: disabled)
     Active: active (running) since Sun 2025-07-06 11:04:54 EDT; 1min 1s ago
 Invocation: 9c943729b242406093e82763cbd2919e
       Docs: man:smbd(8)
             man:samba(7)
             man:smb.conf(5)
   Main PID: 101133 (smbd)
     Status: "smbd: ready to serve connections ... "
      Tasks: 3 (limit: 4512)
     Memory: 23.2M (peak: 23.5M)
        CPU: 256ms
     CGroup: /system.slice/smbd.service
             ├─101133 /usr/sbin/smbd --foreground --no-process-group
             ├─101145 "smbd: notifyd" .
             └─101146 "smbd: cleanupd "

Jul 06 11:04:53 Lior5568 systemd[1]: Starting smbd.service - Samba SMB Daemon ...
Jul 06 11:04:54 Lior5568 systemd[1]: Started smbd.service - Samba SMB Daemon.
```

systemctl status of SSH:

```
┌──(lior5568㉿Lior5568)-[~]
└─$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: disabled)
     Active: active (running) since Sun 2025-07-06 06:50:37 EDT; 4h 21min ago
 Invocation: 1c59f3c1676649cc9c1f1026d81b957e
       Docs: man:sshd(8)
             man:sshd_config(5)
   Main PID: 72088 (sshd)
      Tasks: 1 (limit: 4512)
     Memory: 2.3M (peak: 2.6M)
        CPU: 56ms
     CGroup: /system.slice/ssh.service
             └─72088 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jul 06 06:50:37 Lior5568 systemd[1]: Starting ssh.service - OpenBSD Secure Shell server ...
Jul 06 06:50:37 Lior5568 sshd[72088]: Server listening on 0.0.0.0 port 22.
Jul 06 06:50:37 Lior5568 sshd[72088]: Server listening on :: port 22.
Jul 06 06:50:37 Lior5568 systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
```

systemctl status of vsftpd:

```
┌──(lior5568㉿Lior5568)-[~]
└─$ sudo systemctl status vsftpd
● vsftpd.service - vsftpd FTP server
     Loaded: loaded (/usr/lib/systemd/system/vsftpd.service; enabled; preset: disabled)
     Active: active (running) since Sun 2025-07-06 06:51:18 EDT; 4h 23min ago
 Invocation: 346361be2f164e7ab4404b98c8635e4e
   Main PID: 72677 (vsftpd)
      Tasks: 1 (limit: 4512)
     Memory: 1.1M (peak: 1.8M)
        CPU: 40ms
     CGroup: /system.slice/vsftpd.service
             └─72677 /usr/sbin/vsftpd /etc/vsftpd.conf

Jul 06 06:51:18 Lior5568 systemd[1]: Starting vsftpd.service - vsftpd FTP server...
Jul 06 06:51:18 Lior5568 systemd[1]: Started vsftpd.service - vsftpd FTP server.
```

systemctl status of apache2:

```
└─$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: disabled)
     Active: active (running) since Sun 2025-07-06 06:47:45 EDT; 4h 28min ago
 Invocation: 112b9f1da63e413ab85bfd3c21a11895
       Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 70423 (apache2)
      Tasks: 6 (limit: 4512)
     Memory: 21.3M (peak: 22M)
        CPU: 1.527s
     CGroup: /system.slice/apache2.service
             ├─70423 /usr/sbin/apache2 -k start
             ├─70426 /usr/sbin/apache2 -k start
             ├─70427 /usr/sbin/apache2 -k start
             ├─70428 /usr/sbin/apache2 -k start
             ├─70429 /usr/sbin/apache2 -k start
             └─70430 /usr/sbin/apache2 -k start

Jul 06 06:47:45 Lior5568 systemd[1]: Starting apache2.service - The Apache HTTP Server...
Jul 06 06:47:45 Lior5568 systemd[1]: Started apache2.service - The Apache HTTP Server.
```

Service Name	Purpose	Port	Check Status
apache2	Web server (HTTP)	80	systemctl status apache2
ssh	Secure remote login	22	systemctl status ssh
vsftpd	FTP server	21	systemctl status vsftpd
smbd (samba)	Samba – File and Printer Sharing	445 , 139	systemctl status smbd

💻 Part 6: Bash Scripting Exercise (15 pts)

Task:

Write a Bash script named user_summary.sh that:

1. Accepts a username as an argument.
2. Outputs:
   - User's UID and GID
   - Home directory
   - Shell
   - Last login (use last or who)

📤 **Deliverables:**

- Script file with comments.
- Sample output using an existing username.

📘 Bonus (Optional +5 pts)

- Configure a shared samba folder accessible with a username and password.
- Document steps taken.

The Script

```
  GNU nano 8.4
#!/bin/bash

# Ask user for input if no argument is provided
if [ -z "$1" ]; then
    read -p "Enter a username: " USERNAME
else
    USERNAME=$1
fi

# Check if the user exists in the system
if ! id "$USERNAME" >/dev/null 2>&1; then
    echo "User '$USERNAME' does not exist inside the Kali Linux system."
    exit 1
fi

# Get UID and GID
UID=$(id -u "$USERNAME")
GID=$(id -g "$USERNAME")

# Get home directory and shell
HOME_DIR=$(getent passwd "$USERNAME" | cut -d: -f6)
SHELL=$(getent passwd "$USERNAME" | cut -d: -f7)

# Try to get last login using 'last'
LAST_LOGIN=$(last -n 1 "$USERNAME" | head -n 1)

# If no result, try 'lastlog'
if [ -z "$LAST_LOGIN" ]; then
    LAST_LOGIN=$(lastlog -u "$USERNAME" | tail -n 1)
fi

# Output
echo "User Summary for: $USERNAME"
echo "- - - - - - - - - - - - - - - - - - - - "
echo "UID: $UID"
echo "GID: $GID"
echo "Home Directory: $HOME_DIR"
echo "Shell: $SHELL"
echo "Last Login: $LAST_LOGIN"
```

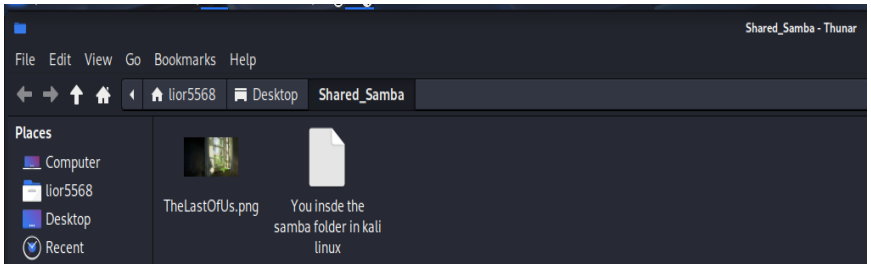
The output of the user alice

```
┌──(lior5568㉿Lior5568)-[~]
└─$ ./user_summary2.sh
Enter a username: alice
./user_summary2.sh: line 17: UID: readonly variable
User Summary for: alice
- - - - - - - - - - - - - - - - - - - -
UID: 1000
GID: 1013
Home Directory: /home/alice
Shell: /bin/bash
Last Login: alice    tty8         :1              Tue Jun 17 09:29 - 10:09  (00:39)
```

Part 6 – Bonus: Samba Shared Folder

The Samba shared folder – in kali

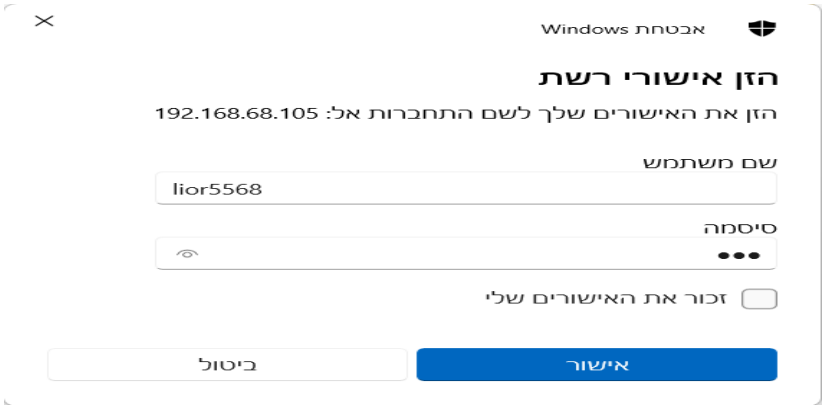
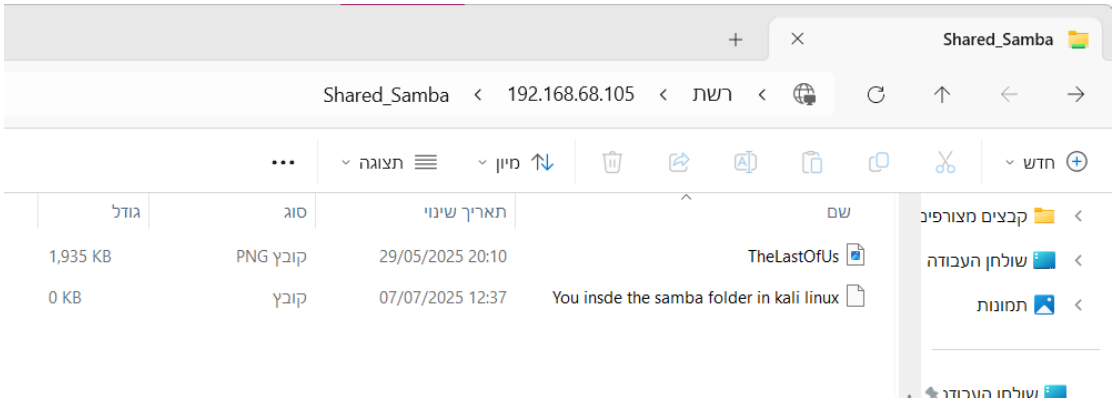
```
┌──(lior5568㉿Lior5568)-[~/Desktop/Shared_Samba]
└─$ ls
TheLastOfUs.png  'You insde the samba folder in kali linux'
```



The Samba script inside the - nano /etc/samba/smb.conf

```
[Shared_Samba]

comment= Shared Folder
path = /home/lior5568/Desktop/Shared_Samba
browseable = yes
read only = no
guest ok = no
writeable = yes
valid users = lior5568
```

Shared samba folder in Windows accessible with a username and password



The shared folder in Windows